

The Flight — Arcade Game 7 Design Document

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Overview

The Flight is a 2–6 player 3D real-time strategy game in which each player is a bird. You start alone on an island with wings, a hunger, and an empty nest. You fish to feed yourself, fish to attract a mate, fish to feed the chicks, and the chicks grow into the colony that fishes, builds, scouts, trades, prays, and fights for you. The sea you fish is Depth's living food web, and the things in it eat birds. Other players' colonies grow on other islands; whoever feeds, grows, and fights best is the flock that owns the sky when the match ends.

What makes it different from Fathoms (Depth's other RTS): you are *in* the game as a unit. The camera is on your back. You fish with your own talons, carry with your own wings, lead attacks with your own beak, and if a shark takes you, your colony loses its leader for a while and you respawn from your nest as a chick-leader with half your stats until you feed yourself back up. The colony is commanded with RTS orders; the leader is flown.

Design pillars

1. **Feed or die.** Every bird eats. A colony that grows faster than its fishing starves back down. The core tension is birds versus fish, all match, and the best players feel it every minute.
2. **The sea is alive and it is not yours.** Fish move, breed, flee, and are eaten; sharks follow the blood your fishers make; the kraken cove and skull island are ecosystems, not spawn points. The food-web engine from Red Tide runs the ocean.
3. **Your bird matters.** Twelve founder species with boosts that bend the whole strategy, plus the fact that the leader is a flyer with a body: altitude, wind, stamina, and a shark's open mouth are things that happen to *you*.
4. **War is elaborate and learnable.** Scouts report, flocks form, altitude and wind decide fights, raids steal eggs, sieges starve islands, bombers and chemists change the rules late. A new player can win a skirmish; a good player can win a siege.

5. **Islands are characters.** A vertical rock, a tropical island, a fishing town, a volcano, a cove with a kraken, an island with giant beasts: each is a different economy, a different defense, and a different risk.

Session. 25–45 minutes, 2–6 players, free-for-all or teams, hosted through the Deep Arcade's shared layer. Score pays arcade tokens; tokens buy bird costumes or spin the loot crate.

Tone. Depth's inked low-poly style, bright and windy: a tropical sea under a brass sky, birds with character, a kraken that is a joke until it isn't. Bird deaths are a feather burst and a splash; chicks starve with a sad peep and a tombstone of shells. Silly names, real strategy.

A note on the founder birds. The twelve are built from the boosts you listed; four are renamed so the game owns its characters: the charisma bird is the Lyrebird, the divinity bird is the Ibis, the egg-thief is the Cuckoo (which is what real cuckoos do), the boom-and-bust bird is the Tycoon, and the bombing-and-chemistry bird is the Alchemist. The mechanics are exactly as described.

You are a bird

The player's bird is the **Founder**: the first bird on the island, the colony's leader, and a unit the player flies directly in third person. Everything the colony can do, the Founder can do better, and the Founder is the only unit the player steers with their hands rather than with orders.

Flight

- **Speed and stamina.** Cruise at 12 m/s; sprint at 20 for as long as stamina lasts (8 s base); stamina refills while gliding. Flapping costs stamina; gliding on an updraft or a tailwind refills it. Carrying a load slows you by its weight.
- **Altitude.** Three bands matter: **low** (under 10 m: fishing, landing, and the band sharks can reach when they breach), **mid** (10–60 m: travel, fights), **high** (60–200 m: scouting range, bombing runs, and the band where wind is strongest). Climbing costs stamina; diving converts altitude into speed (a dive from high reaches 35 m/s and is the fastest thing in the game).
- **Wind.** The map has a wind field that shifts every few minutes and is shown by cloud drift and flag directions on islands. Flying with it is +30% speed; against it, –30%. Thermals rise off tropical islands and volcanoes in the afternoon: free altitude.

- **Landing** anywhere on land, on the nest, on a ship's rigging, on the sea (floating: safe from most things, not from what's below).

Fishing

1. Spot fish from altitude (shadows and ripples; schools are dark patches; a Scout's report marks them on the map).
2. Dive. A dive aimed at the water enters a **strike**: a slow-motion 0.5 s in which you steer the talons onto a fish. Hit chance depends on the fish's size, your talon stat, the dive speed, and the water's clarity.
3. Grab. A fish up to your **carry** stat comes up with you; a heavier one pulls you under for 2 s (vulnerable) and escapes. A Strongbird carries more; a Taloned bird grabs easier.
4. Fly it home. A carried fish is weight (speed down) and scent (sharks and frigatebirds follow). Drop it in the nest or at a **cache** (a storage pile) and it becomes **feed**.
5. Eat. Your own hunger is a bar that drains over a game day; eating a fish from a cache refills it. A starving Founder loses stats, then faints at 0 and is a shark's dinner if it faints over water.

Carrying and building

The Founder carries one thing: a fish, a twig, a shell, a chick (to move it), an egg (yours or someone else's), or, late, a bomb. Nests and structures are built by dropping materials on a site; the Founder can build alone at first and directs Builders later.

Dying

The Founder can be eaten by a predator, killed in a fight, drowned by a load, or caught in a volcano's ash. On death:

- The colony keeps running on its last orders for 60 s, then goes **leaderless**: no new orders, fishers keep fishing, warriors go home, fervour drops.
- The player respawns at the nest as a **chick-leader** after 30 s: half speed, half carry, no fighting, and hungry. Feeding yourself three fish restores the adult Founder. Ten more seconds at full stats and the colony takes orders again.
- With no nest (the island was taken), the player respawns as a chick on a drifting log and must be found by their own colony or land on neutral ground. Three Founder deaths in a match is -10% to everything for the rest of it (the colony's confidence).

What the Founder does that the colony can't

Fly the camera, scout in person, pick the mate (by delivering the courtship fish), start a new colony (by landing on another island and building the first nest), lead a flock (a flock led by the Founder gets +10% speed and morale), carry a bomb, make a trade in person at a fishing town, and pray at the shrine for a fervour burst. The best players are flying between these jobs all match.

The colony loop

A colony is birds, and birds come from eggs, and eggs come from a mate, and all of it comes from fish. The loop is: fish → feed yourself → feed a courtship → a mate → eggs → chicks (fed) → adults (fed) → more fish, more nests, more mates.

A game day

The match runs on a day cycle of about 4 real minutes: dawn (fish rise, the best fishing), day (building, war), dusk (fish rise again), night (predators hunt, scouting is hard, birds sleep unless fervour keeps them up). Chicks age by days.

Hunger and feeding

- Every bird has a hunger bar that empties over one game day. A bird at 0 for half a day **starves** (dies). Chicks empty twice as fast.
- Feed comes from caches. **Feeders** (a role) carry fish from caches to nests; adults fetch their own if no Feeder is assigned. The colony panel shows **feed per day** against **mouths per day**; the gap is the whole economy.
- A fish's feed value is its size (1–6, from the food web): a sardine feeds one chick for a day; a tuna feeds a nest.
- Overfeeding does nothing except make chicks grow faster (+25% at full), so surplus is for growth, trade, or courtship.

Courtship and mates

1. Build a nest (twigs and a site). A nest holds one pair and up to four eggs.
2. Attract a mate: fill the nest's **courtship bowl** with fish (3 fish of feed value 2+ for the first mate; more for each after). A mate arrives from the sea's wild-bird population within a game day, faster with a Lyrebird founder, with a shrine, or on a tropical island.

3. Mates are colony birds that don't take orders but lay eggs: one clutch of 2-4 eggs per two game days while fed, three clutches per mate. Mates can be lost to predators and raids.
4. Each new pair needs its own nest. The colony is capped by nests, and nests are capped by sites (islands have 6-40 sites by size) and twigs.

Eggs and chicks

- An egg hatches after one game day if **warmed** (a mate or any adult sitting on it) and not stolen or eaten. Unwarmed eggs chill and fail after two days.
- A chick eats for two game days, then **fledges** into an adult of a role the player assigns at fledging (fisher, builder, feeder, scout, warrior type, priest, trader). Changing a role later costs a day of retraining.
- Chicks in a nest are a target: cuckoos, raiders, rats on fishing towns, and skull island's beasts all take them. A nest can be guarded by a Watcher.

Growth, and what growth costs

Colony size	Fish per day needed	Nests	What it can do
1 (the Founder)	2	1	Fish, scout, build
5	8	2	Two fishers, a feeder, a scout; the first clutch is the whole plan
15	24	5	A war party of 4, a trader, a priest; a second island is possible
40	70	12	A real colony: bombers, chemists, two flocks; the first island's fishing grounds are exhausted without range
100	180	30	The cap per island; expansion is mandatory; a siege can starve it

Fishing grounds have a yield (fish per day) that falls when overfished and recovers when left; a colony that outgrows its grounds has to fish farther (slower, more predators) or take another island. This is the game's forced expansion.

Starvation

When feed runs out, chicks die first, then the oldest adults, one per hour each, while the colony panel flashes. A starving colony's warriors desert (they go fishing for themselves) and its fervour collapses. A siege works by making this happen.

Second colonies

The Founder (or a Pathfinder bird, late) lands on another island and builds a nest; one mate and two chicks moved there become a colony with its own caches and panel. Colonies share research and the player's orders; they do not share feed unless a **feed line** (Feeders assigned between islands) carries it, which is a supply line and a target.

Resources and the feed economy

Five resources, all carried by birds, all stored in caches on the island. Fish is the one that matters; the others are what a colony is built from and what it trades with.

Resource	Where	Who gathers	What it's for
Fish (feed)	The sea's fishing grounds; fishing town docks (by trade)	Fishers, the Founder	Eating, courtship, chick growth, trade, bribes, and fervour (offerings)
Twigs	Trees on tropical islands and towns; driftwood on every shore; rigging on wrecks	Builders	Nests, caches, perches, hedges, towers, shrines
Shells	Beaches, reef shallows, crab middens	Builders, chicks (a chick's first job)	Nest lining (eggs hatch faster), armor for Tanks, currency at fishing towns
Pearls	Oyster beds (dive deep, slow, predators), the kraken's cove, skull island's shrine	Fishers with Deep Dive research, the Founder	Research, the shrine's offerings, the costliest trades, the Tycoon's boom

Resource	Where	Who gathers	What it's for
Guano	Any colony makes it; more with more birds	Nobody; it accumulates on the island	Fertilizes the island's trees (+twigs), feeds the reef below (+fish near the island), and is the chemistry tree's base material

Caches

A cache is a pile with a capacity (20 fish at first, upgraded by research). Fish in a cache spoil after two game days; a **smokehouse** (research) keeps them for six. Caches are targets: raiders take from them, and a cache near the shore is a gull's buffet.

Feed per day and the colony panel

The panel shows: birds (by role), mouths per day, feed per day (from fishers' average catch), fish in caches (days of food), twigs, shells, pearls, fervour, and the research in progress. The one number to watch is **days of food**: under 1 is an emergency, over 3 is growth.

Fishing grounds

Each ground is a region of sea with a species mix and a yield. Yield = the food web's standing population there. Overfishing a ground drops its yield (and the predators it supports leave for a while, which is sometimes the point); resting it recovers. Grounds near islands are shallow and safe-ish; open-sea grounds are rich and full of things that eat birds; the kraken's cove is the richest water on the map.

Trade

Fish, twigs, shells, and pearls can be traded at fishing towns (an NPC market) and between players (a barter screen: offer, counter, accept). Prices move with supply; a town with a storm coming pays double for fish. Traders carry goods and make the trip; the Sigma bird's colony gets better rates everywhere.

The twelve founder birds

The player picks a founder species in the lobby. It sets the Founder's own stats and gives the whole colony one bend: a bonus, a unique ability or research, and a weakness, so every species plays the match differently.

Founder	Boost	Unique	Weakness	Playstyle
The Taloned (an osprey)	Fishing hit chance +30% for every bird; the Founder's strike window is twice as long	Talon Lock: a held fish can't be dropped by a hit	-10% flight speed	The safest economy; feed-rich and slow
The Strongbird (a sea eagle)	Carry +2 size classes for every bird; can lift size-4 fish from the start	Haul: carry a chick or an egg and a fish at once	Stamina -20%	Big fish early; a tuna feeds a nest and a colony grows fast on few fishers
The Swift (a tern)	Flight speed +25%, dive speed +40% for every bird	Skim: fish without a full dive (small fish only, no vulnerability window)	Carry -1 size class; chicks fledge a day later	Raids, scouting, and fishing far grounds before anyone else; weak in a straight fight
The Beaked (a shrike-gull)	Attack +30%, every warrior type's damage up	Tear: hits bleed, which the sea notices (predators come to the fight)	Fishing hit chance -15%	War early; wins skirmishes; must take an island to feed itself
The Lyrebird	Mates arrive in half the time; clutches +1 egg; fervour +10%	Serenade: the Founder sings at a nest to call a mate instantly, once per day	Warriors -10% HP	The biggest colony; a numbers game; expansion is its weapon

Founder	Boost	Unique	Weakness	Playstyle
The Shadow (a night heron)	Predators spot birds at half range; fishing at night is safe	Dusk Raid: warriors are invisible to Watchers at night	Fervour cannot exceed 80	Fishes the kraken's cove; raids unseen; a thief's colony
The Sigma (a magpie)	Trade rates +40% at towns and in barter; Traders carry double	Market Cornering: once per match, buys a town's whole fish stock at the morning price and sells at the evening price	Fishing hit chance -10%; no faith tree	Wins the economy through towns; a merchant flock that buys its wars
The Ibis	Fervour +50%; priests convert at twice the rate; birds rest half as much under fervour	Pilgrimage: a shrine built on any island claims it without a nest	Trade rates -20%	A tireless colony that works nights; a conversion army; the strongest siege defender
The Cuckoo	Egg theft from the start (others need research); stolen chicks fledge as yours	Brood Parasite: lay an egg in an enemy nest; the enemy feeds it; it fledges as your Scout inside their colony	Your own clutches are 1 egg smaller	Grows on everyone else's eggs; the game's most hated and most fun founder
The Tycoon (a crested cockatoo with a notable crest)	Boom: once per 3 game days, 2 days of +60% feed, build speed, and research, followed by 1 day of -40%	Golden Nest: the first nest produces pearls (1 per day)	No sustained bonus; the bust day is dangerous if timed by an enemy	Timing: boom to expand, survive the bust, boom again; enemies learn the cycle

Founder	Boost	Unique	Weakness	Playstyle
The Alchemist (a hooded crow)	The bombing tree and the chemistry tree research at 3× speed and unlock a day earlier	Guano Works: guano accumulates twice as fast	Fishing +0, fervour −20%	The late game: bombers over enemy islands and stimulant-fed warriors; weak until the trees come online
The Albatross	Stamina +100%; wind penalties halved; scouting range +50%	Long Reach: fishes grounds two map regions away without a rest stop	Nests cost double twigs (big birds)	The explorer: first to every dangerous island, claims the far ones, holds a wide empire thinly

Founder stats (the player's own bird) follow the species: the Taloned has the best strike, the Strongbird the best carry, the Swift the best speed, the Beaked the best attack, the Albatross the best stamina, and so on. The colony inherits the boost, not the Founder's body: a Swift's fishers are faster, not smaller.

Balance intent. Each founder has a window: the Taloned, Strongbird, and Lyrebird are strongest early; the Beaked, Swift, Shadow, and Cuckoo in the mid-game; the Sigma, Ibis, Tycoon, Alchemist, and Albatross late. The phase model from Fathoms is reused to hold them within $\pm 3\%$ overall.

Colony birds

Every colony bird is one of the player's species, assigned a **role** at fledging. Roles are the unit types. Stats are for a size-3 adult at base; founder boosts and research modify them. All birds have hunger, can be eaten, and sleep at night unless fervour keeps them working.

Working roles

Role	HP	Speed	Carry	What it does	Notes
Fisher	60	12	2	Fishes an assigned ground and returns fish to the nearest cache; the economy	A fisher catches 3–6 fish a day by ground; dies to sharks if the ground is bloody
Feeder	60	14	2	Carries fish from caches to nests and to birds on duty; runs feed lines between islands	Without feeders, every bird fetches its own and loses a third of its day
Builder	70	11	3	Gathers twigs and shells; builds and repairs structures	Three builders raise a nest in a game hour
Scout	50	18	0	Flies a patrol route or a target and reports (see Scouting); sees far, fights never	A Scout over an enemy island at high altitude is the first move of every war
Trader	65	13	4 (goods)	Carries goods to towns and other players; haggles	Traders are unarmed and known; killing one is an insult the towns remember
Priest	55	11	1	Tends the shrine; raises fervour; converts wild birds (and, with research, enemy birds)	Needs a shrine and offerings
Chemist	55	11	2	Brews stimulants from guano and pearls at the Works; applies them to flocks	Alchemist founder only early; everyone late
Pathfinder	70	15	1	Founds new colonies without the Founder: lands, builds, and calls a mate	Late research; the empire's expansion unit

Warrior roles

Warriors are the elaborate part. Each has a job in a flock, a counter, and an altitude it likes.

Warrior	HP	Speed	Attack	Role	Strong against	Weak against	Altitude
Skirmisher (a tern-like strike bird)	50	20	12, fast	Quick attacks: hit, climb, hit again; raids caches and nests; harasses fishers	Fishers, Scouts, Bombers	Tanks, Watchers	Any; dives from high

Warrior	HP	Speed	Attack	Role	Strong against	Weak against	Altitude
Tank (a shell-armored gannet)	180	10	10	Takes hits; a wall in the air; escorts Bombers and Traders	Skirmishers, Strikers' first hit	Bombers' blasts, chemistry-stunned	Mid, holds position
Striker (a hawk)	90	14	30, slow	Strong attacks: the killer; drops on Tanks from above	Tanks, Priests, nests	Skirmishers (too fast), Watchers' nets	High; dives
Watcher (an owl)	80	9	8	Island defense: sees stealth, guards nests, throws nets (holds a bird 3 s)	Skirmishers, Cuckoos, night raids	Strikers, bombs	Perched; never leaves the island
Bomber (a pelican with a pouch)	120	9	0 (bomb 60 area)	Carries one bomb from the Works to a target; drops from high	Structures, Tanks, grouped birds	Skirmishers, Watchers' nets	High only
Screamer (a gull)	60	15	5	Morale: a screamer in a flock raises its morale and lowers the enemy's; three together panic Fishers off a ground	Morale, Fishers	Everything with teeth	Mid
Flockmaster (a crow)	90	13	15	Leads a flock without the Founder: holds formation, gives the flock +5% speed	—	Being killed first	Mid
Fisher-of-Birds (a frigatebird, research)	100	17	20	Steals fish from enemy fishers in the air; a pirate	Fishers, Traders	Tanks	Any

Flocks

Warriors are commanded in flocks of 2-12 with a leader (the Founder or a Flockmaster). A flock has a **formation** (see War), a target, an altitude order, and a stance (raid, hold, escort, retreat-when-half). Birds outside a flock fight as individuals and lose.

Upkeep

Every bird is a mouth. Warriors eat 1.5× a fisher. A colony of 15 fishers can feed about 8 warriors at rest and 5 on campaign (flying burns feed). A standing army starves a colony; armies are raised for a war and demobilized to fishing after it, which is the rhythm the game teaches.

Islands

Every island is a different economy and a different defense. Starting islands are safe and fair; the dangerous ones are the map's center and its best ground, if a player can hold them.

Starting islands (one per player, chosen in the lobby or random)

Island	Shape	Nest sites	Economy	Defense	Weakness
The Sea Stack	A vertical rock 120 m tall with ledges	20, all on cliffs	Shells and guano; fish in the deep water around it; few twigs (driftwood only)	The best in the game: attackers must climb, Watchers on ledges see everything, Strikers dive from the summit	No trees; feed lines are vulnerable; a volcano's ash or a storm empties the ledges
The Tropical Island	A green island with a lagoon and a hill	40, in trees and on the beach	Twigs everywhere; lagoon fish (safe, small); thermals in the afternoon	Trees hide nests; a hedge research makes the forest a maze	Open beaches to land on; rats; the lagoon's yield runs out by mid-game

Island	Shape	Nest sites	Economy	Defense	Weakness
The Fishing Town	A human village with docks, boats, a church tower, and nets	25, on roofs, in the tower, on the boats	Trade at the dock (fish for anything), twigs from the woodpile, the fleet's catch to steal	Humans shoo attackers (a Watcher that never sleeps); the tower is a fortress	Humans shoo you too (a tax: 10% of caches); rats take eggs; a Shadow founder's colony is tolerated, a Beaked one is chased
The Atoll	A ring of sand around a lagoon	15 on the ring	The reef's fish (medium, varied); shells and pearls in the lagoon	Nothing to hold; a raid can land anywhere	Everything

Arrangements (the lobby's map option)

- **Archipelago:** starting islands 300–500 m apart with small neutral islets between; dangerous islands in the middle. Fast games, early contact.
- **Safe Distance:** starting islands 900–1,200 m apart with open sea between; the dangerous islands in the center are the only reason to cross. Slow games, big colonies, long wars.
- **Ring:** starting islands around a central kraken cove; every expansion is toward the same water.
- **Chain:** islands in a line; your neighbors are your only neighbors.

Maps are generated with rotational fairness (Fathoms' rules): equal distance to the nearest dangerous island, the nearest town, and the nearest neighbor.

Dangerous islands (neutral; beneficial if held)

Island	What lives there	The danger	The prize	How to hold it
Kraken Cove	A kraken in a sea cave; its arms surface in the cove; squid, big fish, and the pearls it hoards	The kraken grabs any bird under 20 m over the cove; it surfaces fully when blood is in the water; it attacks nests on the cliffs above when the colony is loud (noise)	The richest fishing on the map (3× yield); pearls on the cove floor; the kraken ignores a colony that offers it a fish a day (a priest's job)	Nest on the cliffs above 20 m; fish the cove at dawn and dusk only; keep a priest offering; never bleed in its water
Skull Island	Giant beasts: a great ape that throws rocks at anything flying low, a tribe of giant lizards, a swarm of giant insects, carnivorous plants that eat chicks	Low flight anywhere is a thrown rock; the lizards climb to nests; the plants grow onto nest sites	The island's trees are the best twigs on the map; the beasts' kills leave carcasses (feed); a shrine at the summit gives fervour +20% to whoever holds it	Nest high in the tallest trees; fly in high; feed the ape (it sleeps for a day after a tuna); burn the plants (a bomb)
Volcano Island	Nothing much: hot rock, sulfur vents, a crater lake of strange fish	Eruptions every 6 game days (tremors warn): ash kills chicks and grounds birds for a day; lava takes nests on the slopes	Thermals all day (free altitude for Bombers and Scouts); the crater lake's fish are size 4 and nothing eats birds there; sulfur is the chemistry tree's rare material	Nest on the far side from the crater; move eggs during tremors; harvest sulfur between eruptions

Island	What lives there	The danger	The prize	How to hold it
The Wreck	A drifting hulk that moves slowly across the map	Rats, a ghost crew at night, the hull sinks a little each day	Rigging for twigs, a hold of salted fish (a cache that spoils slowly), the ship's bell (a morale item)	It isn't held; it's raided; it sinks by the late game
The Reef Garden	A coral maze of small fish and eels	Eels take chicks; sharks at the drop-off	Shells in heaps; a cleaning station calms the sharks (Red Tide's rule)	Nest on the one sand cay; keep the cleaners alive

Colonizing

Any island with a nest site can be colonized: the Founder (or a Pathfinder) builds a nest, a mate is called, two chicks are moved, and the island gets its own caches. An island is **held** by whoever has the most nests on it; two players can share one until they don't.

Dangerous islands give their prize to whoever holds them and their danger to everyone.

Weather on islands

Storms ground birds for a game hour, raise the sea (the atoll floods; the stack doesn't), and empty the sky of scouts; fog hides raids; the dry season halves tropical twigs; the kraken surfaces in storms.

The living sea

The ocean is Red Tide's ecosystem engine running under an RTS: species with diets, blood and scent, sound, territory, and population, simulated by the host. Birds are members of the web: they eat fish, and some things eat them. The sea is why fishing is never free and why a war's blood matters.

What's in the water (about 30 species per map)

Tier	Examples	What they mean to a bird
Bait (size 1)	Sardines, anchovies, krill	The lagoon economy: safe, small, endless until overfished
Table fish (size 2–3)	Mackerel, snapper, mullet, squid, flying fish	The main economy; schools move with the day; a Strongbird's first targets
Big fish (size 4)	Tuna, mahi, barracuda, kingfish	A nest's dinner in one trip; need carry 4; they fight the talons (a 2 s struggle; the shark's window)
Things that eat birds	Sharks (size 4–5), the great white, barracuda packs, a giant grouper under the stack, crocodiles at the tropical island's river mouth, the kraken, skull island's lizards and ape, frigatebirds (wild, steal fish)	Attracted by blood from struggles and fights, by splashes (sound), and by a bird on the water; they breach to 10 m
Grazers and cleaners	Turtles, dugongs, cleaner wrasse	Not food (too big or protected); the cleaners calm the sharks near reefs
Scavengers	Crabs, gulls, hagfish	Clean up the blood a fight leaves

The rules birds live by

- **Blood.** A fish that fights on the talons bleeds; a bird that falls bleeds more; a war over water is a feast for sharks, who then hunt the fishers of both sides. Chemists' stimulants make birds bleed less; the Beaked founder's Tear makes enemies bleed more. The sea is a weapon if you can aim it.
- **Splash and noise.** Every dive is a splash the web hears; a ground fished by ten fishers at once draws predators in minutes; spreading fishers across grounds is safer and slower.
- **Yield.** A ground's yield is its standing population; it recovers over days if rested; a kraken's cove recovers overnight because the kraken keeps everything else out.
- **Predator pressure** per ground is shown on the map as a shark icon that darkens with risk; Scouts' reports update it.

- **Breaches.** A shark can take a bird below 10 m; a crocodile below 5 m near shore; the kraken below 20 m over its cove; frigatebirds steal from any carrying bird they catch in the air (Tanks and Watchers drive them off).
- **Night.** Predators hunt harder; fishers rest; the Shadow founder's colony fishes anyway.

The map's monsters

- **The kraken** is a persistent agent: it sleeps unless blood or noise wakes it, grabs low birds with arms, and surfaces fully in storms or when offered nothing for three days. It can be fought (a Bomber campaign and a lot of blood) for the cove's hoard; killing it is a match-defining event and the cove's yield halves without it (nothing keeps the sharks out).
- **The great ape** on skull island throws rocks at low flyers and sleeps when fed. It can't be killed; it can be lured to another player's nests with a trail of fish.
- **The giant grouper** under the sea stack eats anything that falls; it is the stack's moat.

Birds in the web

Wild birds (gulls, frigatebirds, terns, boobies) live on every map, fish the same grounds, steal from caches, and are the pool mates come from. A colony that kills the wild birds near it gets fewer mates. Priests convert wild birds into colony birds (one per game day per priest), which is the Ibis founder's growth engine.

Scouting, the map, and reports

You know what your birds have seen. The map (the **M** key, or the chart button) shows the world as your colony has observed it: islands you've landed on in full, islands you've flown over as silhouettes, grounds your fishers have worked with their yields and predator pressure, and everything else as fog. Enemy colonies appear as what your scouts last reported, with a timestamp.

The map screen

- A hand-drawn chart in the ink style, revealed in patches by every bird's sight. Fog is cloud; old information fades to sepia (a nest count from two days ago is drawn faded, with "seen 2 days ago").

- Layers: islands and sites, fishing grounds (yield and shark icon), wind arrows, enemy sightings (nests, caches, flocks, Watchers), the kraken's last known mood, trade routes, your feed lines, and your flocks' positions and orders.
- Orders can be given on the map (a flock to a target, a fisher group to a ground, a scout to a route) and in the world.
- A **report log** on the side lists every scout report of the match, newest first, clickable to center.

Scouts and reports

A Scout flies a route or to a target at the altitude you set, looks, and comes back (or stays on station until relieved). When it returns within range of a nest or the Founder, it **reports**: a short line and a map update.

Report	What it tells you
Ground	Species mix, yield (abundant / fair / thin / fished out), predator pressure, and which enemy is fishing it
Island	Nest count, cache count, Watcher positions, structures, the island's held-by banner, the state of its hedges and towers
Flock	An enemy flock's size, composition, altitude, and heading
Danger	A kraken awake, an eruption's tremors, a storm front, skull island's ape on the move, a shark pack near your fishers
Opportunity	An unguarded cache, a nest with eggs and no Watcher, a wreck surfacing, a town paying double

A scout at high altitude sees far but counts badly (nests $\pm 30\%$); at mid it counts well but Watchers see it; at low it sees inside hedges and is a target. A scout that dies doesn't report. Two scouts on the same target cross-check (counts become exact).

Watchers and counter-scouting

Watchers see Scouts at mid and low altitude and mark them; a marked Scout's report is known to the enemy (they know you looked). Owls (Watchers) at night see everything but the Shadow founder's birds. Fog and storms blind both sides.

Wild information

Towns sell gossip (what the fleet saw) for fish. The fortune-teller-equivalent is the **old pelican** at every town who, for a pearl, tells you one true thing about the enemy's island. Frigatebirds you drive off drop the fish they stole and a feather that shows who they robbed.

Research: the nine trees

Research happens at the **Roost** (the colony's first structure after the nest) and costs pearls, shells, and days. One research at a time per Roost; a second Roost on a second island runs a second line. Each tree has four tiers; tier 4 is the match-changer. The Alchemist founder runs two of the trees at 3×.

Tree	Tier 1	Tier 2	Tier 3	Tier 4
Nesting	Shell lining (eggs hatch 25% faster)	Big nests (6 eggs)	Cliff nests (sites on vertical rock anywhere)	Hanging colony (nests under overhangs, immune to rock-throws and ash)
Fishing	Schooling eye (fishers see schools from farther)	Deep dive (oyster beds, pearls)	Night fishing (fishers work dusk and dawn safely)	Cooperative fishing (fishers herd a school together: double yield, triple splash)
Flight	Thermal riding (free altitude on tropical and volcanic islands)	Long wings (stamina +50%)	Wind reading (wind penalties halved; the map shows wind a day ahead)	Formation flight (flocks fly 30% faster in a V)

Tree	Tier 1	Tier 2	Tier 3	Tier 4
Caches and craft	Smokehouse (fish keep six days)	Big caches (50)	Shell armor (Tanks +60 HP)	Nets (Watchers' nets hold 5 s; Fishers can net small schools)
War	Flock orders (formations)	Skirmisher and Striker roles	Watchers' towers (sight +50%, nets) and hedges (walls of thorn around nest sites)	Flockmaster and the siege order (a flock blockades a ground)
Bombing (Alchemist 3×)	The Works (makes bombs from guano and sulfur: 1 per day)	Bomber role	Incendiary (burns hedges and nests; skull island's plants)	Blockbuster (one bomb per two days; destroys a Roost or a tower outright)
Chemistry (Alchemist 3×)	The Works (shared with Bombing) and the Chemist role	Stimulant I: Haste (a flock +30% speed for a day; afterward, a day of half stamina)	Stimulant II: Fury (attack +40%, fear 0: the flock won't retreat; bleeds more) and Clot (bleeding halved)	Stimulant III: the Draught (one flock, one day: all stats +50%, immune to morale; then every bird in it sleeps a full day, wherever it is)
Faith	The shrine and the Priest role (fervour)	Offerings (fish for fervour; +10 per big fish)	Conversion (priests turn wild birds, then, at tier 4, enemy birds within sight of the shrine)	Zeal (at fervour 100, the colony works through nights and warriors never rout; fervour drops 30 if the Founder dies)

Tree	Tier 1	Tier 2	Tier 3	Tier 4
Trade and parasitism	Trader role and the market screen	Barter with players (the offer screen)	Egg theft (any Skirmisher can carry an enemy egg home; Cuckoos have it from the start)	Brood parasitism (lay an egg in an enemy nest; Cuckoos from the start) and the Pathfinder role

Costs

Tier 1: 2 pearls, 10 shells, half a day. Tier 2: 5 pearls, 20 shells, a day. Tier 3: 10 pearls, 40 shells, 2 days. Tier 4: 20 pearls, 80 shells, 3 days, and the colony must be 25 birds or more. Pearls are the research bottleneck by design: the oyster beds, the cove, and the trade screen are where research comes from, so a colony with no deep divers and no traders is a colony with no late game.

Order of work the game expects

Nesting 1 and Fishing 1 by day 2; War 1 and 2 by day 4 (or you'll be raided); Faith or Trade by day 5 by temperament; Bombing and Chemistry from day 7; tier 4s from day 10. A 30-minute match is about 8 game days, so most matches see one or two tier 4s, which is the point of them.

War

War in The Flight is fought in three dimensions by birds that get tired, get hungry, and get eaten. The rules below are the strategy system; a player who learns them beats a player with more birds.

The five things that decide an air fight

1. **Altitude.** Height is energy. A flock above its enemy dives for +50% damage on the first strike and can disengage upward; a flock below can only climb (slowly) or run. Strikers win from above and lose from below. The first order in any fight is "get above them".
2. **Wind.** A flock flying downwind is faster and arrives first; a fight against the wind costs stamina every second. Wind shifts every few minutes and is on the map; a flock that attacks into a headwind turning to a tailwind halfway is the classic mistake.

3. **Stamina.** Flapping fights last 20–30 s before a flock is exhausted (half speed, no climb). Diving and gliding cost nothing. The flock that makes the other one flap loses nothing and wins the second engagement.
4. **Morale.** Every flock has morale (0–100): leadership (the Founder +20, a Flockmaster +10), fervour, Screamers, recent wins, and feed set it. At 30 a flock retreats home on its own; at 0 it scatters and is eaten. Killing the leader is –30 at once.
5. **Blood.** Birds that fall into the sea call sharks; a fight over water for more than a minute is a fight over a shark pit, and fishers on both sides will die for days after. Fight over land to keep your sea clean; fight over the enemy's grounds to poison theirs.

Formations (War tier 1)

Formation	Shape	Good for	Weak to
V	A chevron behind the leader	Travel (+speed with Flight 4), arriving together	Being hit from the side; the leader is in front
Wall	Tanks in a line, others behind	Holding an altitude band over an island; escorting Bombers	Strikers diving onto the line's center
Spiral	The flock circles and climbs on a thermal	Gaining altitude before a fight; defending a sea stack	Being caught at the bottom of the spiral
Scatter	Every bird for itself	Skirmisher raids; running	Any organized flock
Hammer	Strikers high, Skirmishers low, Tanks between	The set-piece attack: Skirmishers bait the enemy down, Strikers fall on them	Watchers' nets on the Skirmishers; a Scream
Cover	Tanks over a target (a cache, a Trader, a Bomber)	Escort	Bombs; being outlasted (Tanks tire)

Raids

Small, fast, and about feed, not kills:

- **Cache raid:** Skirmishers take fish from an enemy cache and fly home; a Watcher on the cache nets one; the rest escape. Every fish stolen is a fish you didn't catch and they did.
- **Egg raid:** with Trade and parasitism tier 3 (Cuckoos from the start), a Skirmisher carries an egg home; it hatches yours. The most hated raid in the game and the one that starts most wars.
- **Fisher harassment:** a Skirmisher pair over an enemy ground makes fishers drop their catch and flee; the dropped fish bleed and the sharks come, and the ground is spoiled for a day.
- **Chick snatch:** Strikers take chicks from unguarded nests (skull island's lizards do this to everyone); a chick taken is a mouth you don't feed and a mate they wasted.

Sieges

A siege starves an island:

1. **Blockade** (War tier 4): a flock holds the enemy's fishing grounds; their fishers can't work them without a fight.
2. **Cut the feed lines:** Feeders between islands are unarmed; a Skirmisher pair kills the line, and the second island starves.
3. **Spoil the water:** fight over their grounds; the blood brings sharks that fish for you.
4. **Hold altitude** over the island with a Wall; their Scouts can't leave and their Traders can't come.
5. **Wait.** Chicks die first, then adults, then the warriors desert to fish. A siege takes 2-3 game days and costs the besieger feed too (a flying army eats 1.5×), so a siege is a race between two hungers, and the colony with more fish in caches wins it.

Assaults

Taking an island means landing on it and destroying nests until you hold the most. Defenders have hedges (walls Skirmishers can't pass; Strikers go over; Bombers burn), towers (Watchers with nets and double sight), a shrine's fervour (defenders don't rout), the island's shape (a stack is a fortress; a beach is a landing), and the island's own monsters (skull island's ape throws rocks at the attackers too). A good assault: Scouts first, a Hammer to break the Watchers, Bombers on the hedges, Tanks to hold the beach, Strikers on the nests, Skirmishers on the caches, and a Priest to claim the shrine.

Bombing (Bombing tree)

Bombers carry one bomb from the Works at high altitude and drop it: a 60-damage blast in 8 m that breaks hedges, towers, and nests and scatters flocks. They are slow, loud, and visible from far; they need a Cover escort. Incendiaries burn hedges and the plants on skull island; a Blockbuster destroys a Roost, which stops research and is a declaration of total war. Bombs on the kraken's cove wake the kraken for everyone. The Alchemist founder's colony has bombers by day 4; everyone else by day 8.

Chemistry (Chemistry tree)

Chemists brew stimulants at the Works and dose a flock before it flies: Haste for a raid, Fury and Clot for an assault, the Draught for the match's deciding battle. Every stimulant has a crash; a Draughted flock that wins the battle sleeps a full day in enemy territory, which is the sort of thing a scout should notice. Stimulants also work on fishers (Haste for a fishing race) and the Founder (dangerous: the crash fells the leader).

Defense

Watchers on perches, nets, hedges around nest sites, towers at the landing points, a shrine, a cleaner station on the reef to keep your own sharks calm, and the habit of keeping fish in caches: an island with three days of food is unsiegeable in a 30-minute match.

The war the game expects

Days 1-3: no war (nobody has warriors; raids are theft). Days 3-6: raids and counter-raids, egg theft, fisher harassment, the first fights over the center's grounds. Days 6-10: sieges of a weak neighbor, assaults on dangerous islands, bombers. Day 10+: the deciding assault, usually on whoever holds the kraken's cove. The match ends before anyone can hold everything, which is why it's scored.

Faith, trade, and the boom

Faith and fervour

Fervour is a colony stat from 0 to 100, raised by a shrine, priests, offerings, and the Founder's prayers, lowered by starvation, defeats, the Founder's death, and Shadow founders' cap. What it does:

Fervour	Effect
0–19	Birds rest a full night; warriors rout at morale 40
20–49	Normal
50–79	Birds rest half a night (the colony works late); morale +10
80–99	Birds rest two hours; fishers fish dusk and dawn without research; morale +20
100 (Zeal)	No rest; warriors never rout; priests convert enemy birds in sight of the shrine; a colony at Zeal eats 20% more (tired birds are hungry birds)

The Founder prays at the shrine for a **fervour burst** (+15 for an hour, once a day). The Ibis founder lives here; the Sigma has no faith tree at all.

Trade and barter

- **Towns.** Each fishing town has a dock market with prices for fish, twigs, shells, and pearls that move with supply and events (a storm doubles fish prices; a festival doubles twigs). Traders carry goods there and back. Towns also sell gossip, the old pelican's truth, and, if the colony is liked (a reputation from trade volume and no stolen fish), a **fleet contract**: the town's boats fish a ground for you for a day.
- **Players.** The barter screen: offer any goods, a research (sharing it costs a pearl), a truce for N days, a feed line's passage, or a mate (a bird transferred); the other side counters or accepts. Traders carry the goods; a trade with a player you're at war with is possible and is a story.
- **Reputation.** Each town tracks each player: trade raises it, stealing from the fleet or killing a Trader lowers it; at -50 the town's humans shoo your birds off its roofs; at +50 the town defends your Traders with its own gulls.

The Sigma's cornering

Once per match the Sigma buys a town's whole fish stock at the morning price and sells it back at evening (prices rise through the day); the profit is pearls, which is research, which is why the Sigma wins late.

The Tycoon's boom and bust

The Tycoon's Boom is a button: 2 game days of +60% to feed, build speed, and research, then 1 day of -40% to all three. It's on a 3-day cooldown, so a Tycoon can boom every other day of a long match. Everyone on the map sees the Tycoon's banner change color when the boom starts (and when the bust does), so the bust day is when neighbors attack. The Tycoon's play is to boom into a war or an expansion and have the fish cached before the bust; the counter-play is to scout the banner.

Match structure and victory

Lobby

2-6 players; founder species; starting island (or random); map arrangement; teams or free-for-all; match length (20, 30, 45 minutes); dangerous islands on or off; the kraken's mood (sleepy, normal, hungry); starting season.

Victory

The match ends at the time limit or when one player (or team) holds **two-thirds of all nest sites in use** on the map. At the limit, the highest **colony score** wins:

Score	Points
Living birds	2 each (chicks 1)
Nests	5 each
Islands held	30 each; dangerous islands 80
Fish in caches	1 per 5
Pearls	3 each
Research tiers	10 each; tier 4s 40
Enemy birds killed	1 each; enemy nests destroyed 5; eggs stolen 5
Fervour at 100 at the end	50
The kraken killed	150 (and the cove's yield halves for everyone)
The Founder never died	60

Eliminated players (no nests, no Founder respawn point) watch and can rejoin as a wild bird (an observer who can fish and heckle).

Modes

Mode	Rule
The Flight (standard)	As designed
Flock Together	Teams of 2–3 share research and feed lines; a team's score is summed
Still Waters	No dangerous islands; a longer limit; the pure economy race
Kraken Season	The kraken is awake from the start and roams; whoever kills it wins outright
Nest Egg	Every player starts with 10 birds and no Founder respawn: one life for the leader
Migration	The map's wind is a constant gale that rotates each day; everyone's islands are on the move (the atolls drift)
Solo Season	One player, bots at three difficulties, and the sea

Pacing of a 30-minute match (about 8 game days)

Day 1: fish, nest, mate. Day 2: eggs, Roost, the first scout. Day 3: fledglings, first fishers, War 1. Day 4: raids begin; the center's grounds are contested. Day 5: second islands; sieges start. Day 6–7: bombers, stimulants, the dangerous islands change hands. Day 8: the deciding assault or the score.

Costumes

Arcade tokens buy costumes for the Founder (the player's own bird) and, separately, **colony liveries** (a color and a hat for every bird in the colony, so an army is recognizable from across the map). There are 50 Founder costumes: 10 in the token shop and 40 in the loot crate. Costumes change nothing but the silhouette, and the silhouette is visible to everyone, which in a game about scouting is a choice.

Tokens

10 per match, 1 per 50 score, 5 for each first (first kraken kill, first dangerous island held, first tier 4), 20 for a win.

The token shop (10 Founder costumes)

Costume	Cost	Look
Sea Legs	50	A sailor's cap and a neckerchief
The Lighthouse Keeper	80	A lantern on a chain; glows at night (and is seen at night)
The Admiral	100	A bicorne and epaulettes
The Pearl Diver	100	Goggles and a shell necklace
The Fisherman	120	A hat of hooks; wild gulls follow you
The Scarecrow	120	Straw and a burlap head; crows won't steal from you (a joke; they will)
The Pirate	150	An eyepatch, a bandana, a wooden leg
The Monk	150	A cowl; your shrine's bells ring when you land
The Kraken's Friend	200	A tentacle scarf; the kraken makes a sound when you fly over the cove
The Captain Nemo	250	The coat, the beard, a tiny periscope hat

Colony liveries (token shop)

Six color sets (storm grey, sunset, lagoon, ash, gold, red tide) at 60 each and six hats for the whole colony (caps, helmets, crowns, bandanas, tiny top hats, shells) at 80 each.

The loot crate (42 Founder costumes)

One token, one crate, from a nest of eggs in the lobby: tap an egg, it hatches a costume. Duplicates hatch a feather and nothing else. Odds: Common 60% (15 costumes, 4% each), Rare 25% (10, 2.5%), Super Rare 12% (10, 1.2%), Special 3% (5, 0.6%).

Tier	Costumes
Common (15)	The Rubber Duck, the Chicken, the Penguin (you waddle on land), the Puffin Beak, the Flamingo (pink, one leg when perched), the Parrot (repeats your last order aloud), the Pigeon (a city pigeon, unbothered), the Turkey, the Owl Glasses, the Swan Neck, the Pelican Pouch (purely cosmetic; you can't carry more), the Toucan Beak, the Peacock Tail (drag when landing), the Seagull with a Chip, the Crow with a Shiny Thing
Rare (11)	The Dodo (extinct, undaunted), the Albatross Cape, the Phoenix (ember particles), the Kiwi (a tiny flightless suit that flies anyway), the Stork with a Bundle, the Hummingbird (a blur), the Vulture's Ruff, the Heron Hat, the Cuckoo Clock (a clock on your chest that strikes the game hour), the Carrier Pigeon (a tiny scroll; your reports arrive with a stamp), German Goose (A goose with a black armband)
Super Rare (10)	The Roc (a giant shadow; your silhouette is three times its size, which everyone sees), the Thunderbird (lightning on dive), the Griffin (a lion's rear), the Kraken Costume (eight dangling arms), the Sand Worm Puppet (from A Night Off's crate, for the ambitious), the Diving Bell Helmet, the Fish Costume (you look like your own lunch; sharks are not fooled), the Nautilus (the submarine with wings), the Goat with Wings (the goat), the Depth Diver (the brass helmet from Red Tide)
Special (6)	The Golden Egg (you are an egg with wings; chicks follow you), the Great Ape's Hat (skull island's ape waves), the Sun God's Mask (from the Island; glows at dusk), the Matriarch (an orca costume; sharks give you room), the Founder's Crown (a crown that shows your best match score to everyone, forever), donald the quack (styled after donald trump, particularly the swoop hair)

The crate, the shop, and the odds are data (`flight_costumes.json`), as in A Night Off, and the two games' crates share the arcade token. Costumes earned in one arcade game stay in that game; the Sand Worm Puppet is the exception and the joke.

Presentation, technology, and sound

Look

- Depth's inked low-poly in daylight: a bright sea, white cliffs, green islands, brass-and-sail fishing towns, a sky that goes gold at dusk and ink at night. The ink outline is thinner than Red Tide's (birds are small and far); silhouettes do the work.
- **Birds** are rigged from one body plan with species parameters (wingspan, beak, tail, crest, leg length, a three-tone palette per founder species, a livery color per player). Wings are two-bone rigs with feather quads that spread and fold; Chains for tail feathers, crests, and carried fish. Twelve founder species plus the wild birds and the warrior silhouettes (Tanks are rounder, Strikers are hawks, Watchers are owls, Bombers are pelicans).
- **Flight animation** is physical: flaps on climb, glides on level flight, tucked wings on a dive, a flare on landing, buffeting in wind. A tired bird's flaps are visibly heavier.
- **Fish, sharks, the kraken, and skull island's beasts** reuse Red Tide's `CreatureBuilder` body plans and species records.
- **Islands** are generated from a type's rules (the sea stack is a lathed column with ledges; the tropical island is a noise heightmap with trees; the town is a kit of docks, roofs, a tower; the volcano a cone with a lake) and decorated by habitat rules. The map is 2–3 km across.
- **Weather** is a wind field (shown by clouds, flags, and spray), thermals (visible as shimmer), storms (a front that crosses the map), and fog.

Camera and control

Third person behind the Founder for flight; a free RTS camera (the C key) for orders; the map (M) for the chart. Orders: box-select birds, right-click a ground, nest, cache, or target; flock commands on a bar (formation, altitude, stance); role assignment at fledging on the nest panel. Controller supported (the Founder flies well on sticks).

Simulation and networking

- The host runs the ecosystem (Red Tide's engine), the bird AI (schedules, hunger, roles, flocks), and the economy. About 600 birds and 2,000 fish at full rate near players; population counters elsewhere.
- Host-authoritative through the shared arcade layer: the Founder's flight is predicted locally; orders are commands; everything else is state at 20 Hz with interest management by distance and fog. Six players is the design limit for the bird count.
- Disconnects: the colony runs on its last orders for 2 minutes, then an AI takes it (a cautious one: it fishes and defends).

Sound

- The sea and the wind are the bed: surf by island type, wind rising with altitude, rigging on the town's boats, the cove's slow breathing when the kraken sleeps.
- Birds have voices per species (a cry, a call, a chick's peep) and the colony has a chorus that swells with its size: a colony of 100 is loud, which is how a scout hears one before seeing it.
- Fishing: the dive's whistle, the strike's splash, the struggle, the fish slapping the nest. Feed shortages are heard as hungry peeps before the panel flashes.
- War: wingbeats in formation, the Striker's dive shriek, the net's whip, the bomb's thud and the hedge burning, the Screamer's scream, a flock routing (a scatter of cries), the kraken's roar when it surfaces.
- Reports arrive as a call and a chalk scribble; the map is a paper rustle.
- Music: a windy brass-and-strings theme that layers with colony size and a war motif that enters when flocks engage; a dawn chorus and a dusk hush each game day; the kraken has a theme nobody wants to hear.
- Founders have a signature call each (the Lyrebird mimics the last sound it heard; the Screamer-bird's is the Screamer's).

Build order, data, tests, and open questions

Data files

File	Holds
<code>flight_founders.json</code>	The twelve species: Founder stats, colony boost, unique, weakness
<code>flight_roles.json</code>	Working and warrior roles: stats, upkeep, counters, altitude
<code>flight_islands.json</code>	Island types with generation rules, sites, resources, monsters; arrangements and fairness rules
<code>flight_sea.json</code>	The species list and diet matrix (Red Tide's format), grounds, predator rules, the kraken's script
<code>flight_research.json</code>	The nine trees, costs, effects
<code>flight_war.json</code>	Formations, altitude and wind modifiers, morale, stimulants, bombs
<code>flight_economy.json</code>	Hunger rates, feed values, cache sizes, spoilage, town prices, the Tycoon's boom
<code>flight_scoring.json</code>	Victory, score, modes
<code>flight_costumes.json</code>	Shop, liveries, crate odds

Build order

Stage	Work	Gate
1	Flight: the Founder's body, wind, stamina, altitude, the dive and strike, carrying; one tropical island and a lagoon of fish on Red Tide's engine	Fishing feels good alone for ten minutes
2	The colony loop: hunger, nests, mates, eggs, chicks, roles (fisher, feeder, builder), caches, the colony panel, starvation	A bot colony grows to 40 and starves if it outfishes its lagoon

Stage	Work	Gate
3	Islands: all four starting types, the generator, the arrangements; scouts, fog, the map, reports	A scout report updates the map correctly from each altitude
4	War: formations, altitude and wind rules, morale, the warrior roles, raids, Watchers, hedges, towers	Two bot colonies fight and the higher, downwind flock wins
5	Multiplayer through the shared layer; six founders in a match on LAN	Six players finish a 30-minute match
6	Research trees, faith, trade, towns, barter; all twelve founders	Every founder's window shows in <code>--flight-sim</code>
7	Dangerous islands: the kraken, skull island, the volcano, the wreck, the reef; sieges and assaults; bombing and chemistry	A bot takes the cove and holds it through a siege
8	Costumes, crate, liveries; sound; internet play; the balance pass	All targets met or logged

Tests

- `--flight-sim <founder> <island> <days>` runs a bot colony and reports birds, feed per day, days of food, deaths by cause, research reached.
- `--flight-war <scenario>` runs scripted fights (altitude, wind, formation, stimulants) and asserts the five rules.
- `--flight-fair <seed>` asserts the map's fairness rules.
- `--eco-sim flight <minutes>` (Red Tide's tool) for the sea.

Balance targets

- A careful solo colony reaches 40 birds by day 6 and 100 by day 10 on a tropical island; 30 and 70 on a sea stack.
- A fisher dies to predators once per 15 fishing-days on a safe ground, once per 4 on the cove.
- The twelve founders within $\pm 3\%$ overall with distinct windows (the Fathoms phase model).

- A siege starves a colony with one day of food in 2 game days; with three days of food, never within a 30-minute match.
- The kraken is killed in about one match in ten.

Open questions for the designer

- ☐ Should the Founder be able to die permanently in standard mode (as in Nest Egg), or always respawn as written?
- ☐ Is the kraken killable in every match, or only in Kraken Season?
- ☐ Should colonies share feed automatically across islands, or only by feed lines as written (the siege target)?
- ☐ How many wild-bird mates should the map hold: unlimited, or a pool that the players compete for?

Expansion: the long match

Everything below extends the design above for a 30–60 minute match with more to choose and more to master, while keeping each system small enough to explain in a sentence. The rule for every addition: it is one choice with a clear trade-off, it is visible to other players, and it can be ignored by a new player without losing on day one.

The seasons

A long match runs 12–16 game days through four seasons of 3–4 days each. Seasons change the sea, the wind, and what pays, so the match has acts without a tech ladder.

Season	Days	The sea	The wind	What it rewards	The event
Spring	1–3	Fish rise to the shallows; lagoons are full; mates are plentiful	Light, shifting	Growth: nests, clutches, the first fishers	The Spawning: every ground doubles for a day; the sharks sleep
Summer	4–7	Grounds thin near islands; the blue is rich; thermals all day	Steady trades (predictable)	Expansion, trade, the first wars; Bombers fly cheap on thermals	The Tuna Run: a school of size-4 fish crosses the map in a day; whoever follows it eats for a week

Season	Days	The sea	The wind	What it rewards	The event
Autumn	8–11	Storms; grounds shift; the kraken wakes more	Gales from one quarter for a day at a time	Caches (the smokehouse matters), sieges, raids under fog	The Migration: every wild bird on the map flies through; mates are free for a day, and so is conversion
Winter	12–16	Fish go deep (Deep Dive research pays); the lagoons freeze on northern maps; ice floes appear	Cold winds; stamina costs rise	Holding what you have; the deciding war; the score	The Long Night: one game day of darkness; Watchers and Shadow colonies own it

The lobby picks match length by seasons: Spring and Summer only (30 min), three seasons (45), all four (60). Scores at the end of a four-season match weigh Winter's holdings double, so late colonies matter.

Decrees

At dawn every game day the player picks one **decree** from three offered (drawn from a deck of 24, no repeats in a match). A decree lasts the day. It's one click, it shapes the day, and the deck is where a match's variety lives.

Decree	Effect for the day	Trade-off
Full Nets	Fishers +30% catch	Twice the splash: predators +50%
Quiet Wings	No splash, no noise; raids are silent	Fishers –20%
Feast Day	Chicks grow a full day in half	Caches empty 2× faster
Hatching Moon	Eggs hatch today regardless of warmth	No new clutches tomorrow
Call to Arms	Every adult becomes a warrior for the day (Skirmisher stats)	Nobody fishes; tomorrow they're tired (–20%)
Builders' Day	Build speed 2×; twigs cost half	Fervour –10
Deep Dive	Pearls 2× from oyster beds	Divers can't fish

Decree	Effect for the day	Trade-off
Market Day	Trade rates +30% at every town	Traders are known: pirates target them
Sermon	Fervour +20	Priests can't convert
Scouts Aloft	Scouting range 2×; reports are exact	Scouts can't report until dusk
Egg Watch	Nests can't be raided; cuckoos fail	Watchers can't leave nests
Open Skies	Flocks fly 30% faster	Formation bonuses off
Offerings	The kraken, the ape, and the pirates ignore you	Costs 10% of caches
Brood Day	Mates lay an extra egg	Mates are vulnerable (no warming by others)
Guano Harvest	Guano 3×	The island smells: wild mates avoid it today
Lighthouse	Your island is visible to everyone; traders and mates come from farther	Everyone knows where you are
Fog Bank	Your island is hidden from scouts	Your own scouts are blind
Tithe	Towns pay you 1 pearl per 20 fish traded	Reputation -5 at every town
Mutiny Watch	Warriors can't desert even if starving	Morale -10
Thermal Day	Free altitude over your islands	Bombers everywhere can use it, including theirs
Salvage	Every corpse on your grounds is feed	Scavengers (and their predators) crowd your grounds
Grudge	+20% damage against the player who last raided you	-20% against everyone else
Hospitality	Any wild bird that lands joins you	Any enemy bird that lands is hidden among yours

Decree	Effect for the day	Trade-off
Day of Rest	No decree; fervour +5 and birds heal	—

The Tycoon's Boom replaces the decree on boom days.

Founder perks

The Founder levels three times a match, at days 3, 7, and 11, and picks one perk from three offered. Perks are the player's own build and stack with the species.

Perk	Effect
Broad Wings	Founder stamina +50%
Hooked Beak	Founder attack +30%; the flock the Founder leads +10%
Sharp Eyes	Founder scouts like a Scout at any altitude
Iron Talons	Founder carry +1 size class
Loud Voice	Flocks the Founder leads never rout
Mother's Instinct	Chicks within sight of the Founder grow 25% faster
Pearl Diver	The Founder dives oyster beds without research
Weather Sense	Wind shown a day ahead; storms warned a day ahead
Nine Lives	One free respawn without the chick-leader penalty
Thief's Eye	The Founder sees enemy caches through walls
Diplomat	Barter offers from the Founder in person are +20%
Old Salt	The Founder's reports include the next day's fish movement

Six more founders (eighteen in all)

Founder	Boost	Unique	Weakness	Playstyle
The Gannet	Plunge: every bird's dive from high altitude is a 50% stronger strike, on fish or on enemies	Plunge Strike: the Founder's dive from above 60 m hits a bird or a nest for triple damage	Can't fish from low altitude at all (must climb first)	A vertical colony: everything happens from the top; Strikers and fishers are the same bird
The Pelican	Carry +3 size classes for Feeders; caches hold double	Pouch: the Founder carries three fish at once	Speed -15% for the whole colony	The logistics colony: feed lines that can't be cut by starving them; sieges fail against it
The Frigatebird	Every warrior can steal a carried fish in the air (Fisher-of-Birds for all)	Piracy: the Founder's colony eats what other colonies catch; your fishers may fish enemy grounds unseen	Can't land on water (drowns); nests only on cliffs and trees	A pirate colony that lives off its neighbors and needs no grounds of its own
The Penguin	Birds swim: fishers dive and hunt underwater (no predator breaches; sharks are a fight, not a death); walking is 2×	No flight: the colony's birds don't fly; they swim and walk; islands are reached by sea; the Founder flies (it's the one that can)	No altitude, no bombers, no Strikers' dives; Watchers see nothing above them	The underwater colony: immune to air raids except from above, deadly below, slow to expand

Founder	Boost	Unique	Weakness	Playstyle
The Owl	Night is day: the colony works and fights at night without fervour; Watchers for everyone are free	Silent Wings: the Founder and the flocks it leads make no noise	Daylight: -20% to everything from dawn to dusk	The night colony: raids, scouting, and sieges after dark; sleeps through noon
The Phoenix (a crested firebird)	Chicks that would starve instead fledge at half size (a colony that never fully dies)	Rebirth: the Founder's death respawns instantly at full stats, once per season	Fervour decays 2×; the colony needs a shrine to hold any at all	The comeback colony: hard to finish, easy to bully, dangerous in Winter

More warriors and workers

New warrior roles

Warrior	HP	Speed	Attack	Role	Strong against	Weak against
Plunger (a gannet)	70	13	40 from a dive, 8 otherwise	The dive-bomber: climbs, then falls on one target; useless at level flight	Tanks, Watchers, nests, ships	Being caught climbing; Screamers
Swallow (a swarm unit; trained in fives)	20 each	22	3 each	Cheap, fast, and many: screens, harassment, bait for Hammers; a swarm counts as one bird for feed	Bombers (they can't aim through them), Scouts	Everything with a beak; one Tank

Warrior	HP	Speed	Attack	Role	Strong against	Weak against
Mimic (a mockingbird)	60	15	10	Wears the enemy's livery: flies with their flocks, reports their orders, and turns the Watchers' nets on them	Scouts, Watchers, the enemy's trust	Owls (they see livery through feathers); a Mimic revealed is killed by both sides
Nurse (a gull with a pouch)	70	12	5	Feeds a flock in the field (carries 4 fish) and heals 10 HP a minute to birds within 5 m	A siege's hunger; attrition	Anything that targets it first
Ferrier (a stork)	90	12	2 chicks or eggs	Moves eggs and chicks between islands; founds colonies with a Pathfinder faster	Starvation (it moves mouths to food)	Skirmishers (it can't fight back)
Lancer (a heron)	80	11	25, reach 3 m	Stabs from a perch or hovering; holds a corridor on a stack or a hedge line	Skirmishers in corridors, Swallows	Being out-ranged by Bombers; Plungers
Harrier (a hawk)	85	16	20	The counter-raider: hunts Skirmishers and Fisher-of-Birds over your grounds; patrols on its own	Skirmishers, Frigatebirds, Mimics	Tanks, Strikers from above
Drummer (a grouse)	60	10	5	Morale for the island: perched Drummers raise defenders' morale +15 and make Screamers useless	Screamers, routs	Bombs; it never flies

New working roles

Role	What it does	Why you'd want it
Diver (a cormorant)	Fishes underwater: pearls and deep fish without research; sharks are a fight	The pearl economy without the Fishing tree
Gardener (a parrot)	Plants twig trees on bare islands (sea stacks grow their own twigs after three days) and seeds reef in lagoons (+fish)	Turns a poor island rich over a season
Keeper (a jay)	Runs caches: spoilage halved, counts exact, and stolen fish are tracked (the thief's livery is reported)	Long matches; sieges
Teacher (an old crow)	Chicks within its sight fledge a day early and may be assigned any role, including warrior types usually gated by research	A faster army without the War tree
Herald (a tern)	Carries barter offers and truce flags to other players in person; a Herald in flight can't be attacked without a reputation hit at every town	Diplomacy (below)
Augur (a raven)	Reads the sea: reports where the next day's fish will be and when the next storm comes	The Weather Sense perk for a colony

Veterans and names

A warrior that survives three fights becomes a **veteran**: it gets a name (from a list, or typed), +15% to its stats, a feather marker, and a trait (Fearless: never routs; Lucky: survives one killing blow; Keen: sees stealth; Greedy: steals a fish from every kill; Loyal: follows the Founder anywhere). Veterans are what a player cares about by Autumn, and losing one is the game's grief.

Mates with traits

Every mate that arrives has one visible trait that its chicks inherit as a small bonus: Big Eggs (+10 HP to chicks' adult form), Quick (+5% speed), Hardy (+10% HP), Keen (+scouting), Fierce (+5% attack), Fertile (clutch +1). Courtship can be picky: a nest's bowl filled with a specific fish (the mate's favorite is shown) attracts a mate with the trait you want. Breeding a colony for traits over a long match is the Lyrebird's game and anyone's hobby.

More islands

Six more starting islands (ten in all)

Island	Shape	Sites	Economy	Defense	Weakness
The Iceberg (Winter maps)	A floating berg with ledges and a cave	15	Deep fish and krill under it; no twigs (ice nests need shells)	Attackers can't land on sheer ice; the cave is a fortress	It melts in Summer (sites lost by the day) and drifts with the wind: your island moves
The Lighthouse Rock	A rock with a working lighthouse and a keeper	18, on the lamp gallery and the rocks	The keeper trades fish for lamp oil (a pearl substitute); the beam reveals the sea at night	The beam blinds night raiders; the keeper shoos pirates	Everyone can see your island from anywhere at night
The Shipwreck Island	A galleon broken on a reef	22, in rigging, the hold, the stern cabin	Twigs from the rigging; a hold of salted fish; the ship's bell (morale)	Three decks of corridors; cannon ports as Watcher nests	The wreck settles: the hold floods by Autumn; rats
The Mangrove	A tangle of roots over shallow water	35, in the roots	Crabs and shellfish under the roots (crustacean feed for Divers); endless twigs	Nothing can land in the roots but birds that nest there; crocodiles under the water keep sharks out	Crocodiles under the water; no thermals; Bombers burn mangroves fast
The Kelp Raft	A floating mat of kelp the size of an island	20, on the mat	Fish shelter under it (a lagoon that moves); kelp is twig-grade	It drifts: raiders have to find it; it hides under fog	It drifts into things: the cove, a storm, another player's grounds

Island	Shape	Sites	Economy	Defense	Weakness
The Cliff Town	A human town on a cliff face with a harbor below	30, on balconies and the cliff	Two markets (the harbor's and the town's), the best trade on the map	The cliff is a stack and the town is a town	Two sets of humans to keep happy; the cliff's gulls are a wild colony that competes

Six more dangerous islands (eleven in all)

Island	What's there	The danger	The prize	How to hold it
Iron Island	A naval fort with cannons and a garrison	Cannons fire at any flock over 6 birds; marines shoot nests; the fort's dogs	Powder and shot: the bombing tree at half cost; the fort's stores of salt fish; its flag (a trade item every town pays for)	Nest inside the walls (the garrison tolerates small colonies); never fly in flocks over it; feed the dogs
The Whale	A living island: a sleeping great whale with a colony's worth of barnacles	It dives once a season (everything on it is in the water); it surfaces somewhere else	Krill in its wake (endless size-1 feed); the whale fights the kraken if they meet	Nest on the back; move eggs before the dive (an Augur knows when)
Siren Rocks	Rocks where the Sirens from the Weeds sing	Birds within the song fly to the rocks and sit (lost for a day) unless deaf (Owls, Drummers' sound)	The Sirens trade: a mate of any trait for a pearl; fervour +30 to a shrine built there	Nest upwind of the song; keep Drummers on the rocks

Island	What's there	The danger	The prize	How to hold it
The Maelstrom	A whirlpool a hundred meters wide with rocks at its center	Anything below 30 m over it is pulled in; the rocks at the center are reachable only on a dead-calm day	Everything the sea swallows ends up on the rocks: pearls, wreckage, salt fish, relics	Nest on the center rocks after reaching them; you're untouchable by raids and you can't leave in a storm
The Ghost Ship	A drifting wreck crewed by the Drowned at night	The Drowned take eggs; the ship rams islands it drifts into	Relics in the captain's cabin (below); the ship's cannon can be fired at a neighbor once a season	It isn't held; it's boarded by day and fled by night
Bird Island	A wild mega-colony of ten thousand gulls, boobies, and terns	The wild birds mob anything that lands and eat anything smaller than a tern	Mates without limit; conversion at 10× for an Ibis; guano enough for every chemist on the map	Priests at the edges convert it a flock at a time; an Ibis founder can take the whole island by Autumn; everyone else trades at its edge

Neutral factions

Three neutral powers live on every map and deal with every player by reputation, like Fathoms' pirates.

Faction	Who	What they do	How to deal with them
The Frigate Pirates	A roving flock of wild frigatebirds with a captain	Steal carried fish and cache fish; sell what they steal at a floating market (the Nest of Thieves) that moves each day; can be hired (fish) to raid a named player	Pay them, hire them, or kill the captain (the flock scatters for a season); a Frigatebird founder is their cousin and pays half
The Fishing Fleet	Human boats from the towns	Fish the grounds (competition), drop chum (free feed, sharks), cast nets (a flying bird can be netted at low altitude), and hire birds: a contract pays fish for a flock that drives schools toward the boats	Reputation at the towns; the Sigma gets contracts first; a Frigatebird colony that robs the fleet is hunted by its gulls
The Grey Wings	A clan of wild birds of prey (eagles, hawks) on the map's highest crag	Hunt chicks and lone fishers; can be paid tribute (fish) for a day's peace; a shrine on their crag (Pilgrimage) makes them yours	Tribute, Watchers, or war: killing the Grey Wings' eagle is a relic and a headline

Great events

Beyond the seasons' events, the map rolls one great event per match from this list, announced a day ahead by the Augur (or not at all without one):

- **The Red Tide:** a bloom kills a ground's fish and poisons feed for a day; caches with Keepers are safe.
- **The Kraken's Walk:** the kraken leaves its cove and crosses the map, eating islands' nests below 20 m; the cove is free for a day.
- **The Great Storm:** two days of gales; every drifting island moves; ships wreck (new wrecks to salvage); the lighthouse goes out.
- **The Eclipse:** a day of night in Summer; owls and Shadows own it.

- **The Wreck of the Treasure Ship:** a ship sinks near a random player with a hold of pearls; everyone comes.
- **The Plague:** a sickness spreads through crowded colonies (over 60 birds); the cure is the chemistry tree's tier 1 or a Nurse per 20 birds.
- **The Calm:** no wind for a day; thermals die; Bombers are grounded; everyone's fishers reach everything.
- **The Visitor:** a legendary bird (below) lands on the map and can be recruited by whoever reaches it first with the right gift.

Relics and legendary birds

Relics are single items found on the Maelstrom's rocks, the Ghost Ship, the Whale's back, Iron Island's magazine, and the Grey Wings' crag: a colony holds at most three, at the shrine. Each is one rule: the Ship's Bell (morale +10 everywhere), the Lamp Lens (scouting range 2×), the Kraken's Beak (your flocks ignore the kraken), the Golden Egg (one mate's clutch doubles), the Fort's Flag (every town's rates +20%), the Eagle's Feather (Strikers +20%), the Siren's Shell (mates arrive on call), the Keeper's Logbook (every ground's yield known). Relics can be stolen by raids (the shrine is a target) and traded.

Legendary birds are neutral heroes: one per map, found by the Visitor event or at the end of a quest line (a town's old pelican sets it). Recruit one and it joins your colony as a named unit with a rule: the Albatross of the South (your flocks never tire over water), the Fisher King (a kingfisher; every ground it fishes doubles for a day), the Old Owl (every Watcher sees everything), the Phoenix Chick (a chick that fledges into your founder species' best warrior), the Dodo (does nothing, worth 200 score if alive at the end, and everyone wants to eat it).

Diplomacy, lightly

Six players need a way to talk that isn't a treaty screen. Diplomacy is four things, each one click, each public:

- **Truce** (a day, renewable): neither side's warriors attack the other's birds or nests; fishing the other's grounds is allowed; breaking it costs the breaker 20 fervour and a town reputation hit, and everyone sees the broken flag.
- **Feed pact:** a feed line between two colonies; both panels show it; cutting it is an act of war against both.

- **Bounty:** post fish on another player's Founder; anyone who kills it collects; the posted player sees the bounty and who posted it.
- **Flock loan:** lend a flock to another player for a day for fish; the flock fights under their orders and comes home.

Heralds carry the offers; a Herald in flight can't be attacked without a reputation hit. In team modes the four are automatic between teammates.

Fishing mastery

Fishing has techniques, each a different dive and a different risk, learned by using them; they're what a player gets better at over runs.

Technique	How	Best for	Risk
Plunge	A straight dive from above 30 m; the strike window is short and strong	Big fish; the Gannet's everything	Sharks see the splash; a miss is a 2 s recovery underwater
Skim	Fly along the surface and snatch	Schools of small fish; the Swift	Low altitude over a long line: breaches
Hover-strike	Hold position over a spot and drop	Precision on one fish; crabs and shallows	Stamina; frigatebirds steal from a hovering bird
Drive	Several fishers push a school against a shore or a reef, then strike together	Yield 2× with Cooperative Fishing	Triple splash; the whole shark population comes
Deep dive	Swim down (Divers, Penguins, Deep Dive research)	Pearls, deep fish, the kraken's floor	A fight instead of a flight if something's there
Night fishing	Fish the dusk and dawn rises	Twice the fish per trip	The owls and the Shadows are the only ones it's safe for

A fisher assigned a ground also takes a technique; the panel shows the catch rate and the loss rate per technique per ground, which is the information a returning player reads first.

Nest styles and structures

Nests come in styles, chosen per nest, each a small bet:

Style	Cost	Effect
Cup (default)	10 twigs	—
Platform	20 twigs	Holds 6 eggs; visible from far
Burrow	5 twigs, sand or soil	Hidden from scouts; floods in storms
Hanging	15 twigs, a branch or overhang	Immune to rock-throws and ground raids; sways: Watchers can't perch
Mud	10 twigs, water nearby	Fireproof; dissolves in two days of rain
Cliff ledge	8 twigs, 10 shells	The stack's nest; nothing climbs to it
Floating	25 twigs	On water; immune to everything on land; sharks

Structures beyond nests: caches, the Roost, the shrine, the Works, towers, hedges, perches (a Watcher's post; +sight), smokehouses, the Lookout (a tall perch that doubles a Scout's report range), the Rookery (a nursery: chicks there are warmed by any adult and fledge together), the Beacon (a signal fire: calls every flock home at once), and the Monument (a late, expensive pile of shells that is score and nothing else, which is the kind of thing people build at the end of a long match).

The long match: pacing and scoring

Pacing of a 60-minute match (16 game days, four seasons)

Days	Season	What's happening
1-3	Spring	Nests, mates, the Spawning; decrees begin; the first Founder perk
4-7	Summer	Expansion to second islands; the Tuna Run; trade and the pirates; the first raids; dangerous islands probed

Days	Season	What's happening
8-11	Autumn	Storms; the Migration; sieges; relics found; veterans named; the second perk; the great event
12-16	Winter	The Long Night; the deciding war; the last perk; monuments; the score, with Winter's holdings doubled

Additions to the score

Score	Points
Veterans alive	10 each
Relics held	60 each
A legendary bird alive	100 (the Dodo 200)
Monuments	30 each
Decrees used (variety)	5 per distinct decree, max 60
Truces kept to the end	20 each
Winter holdings	Islands and nests count double

The mastery map

What a player learns over repeated runs, in the order they tend to learn it; it's also the order the game's hints appear:

1. Feed before you grow (days of food).
2. Where the fish are by season, and which technique each ground wants.
3. Which founder fits which island and which friends.
4. Altitude, wind, and stamina in a fight; get above them.
5. Decrees as a daily plan, not a daily gift.
6. Raids are theft: eggs, fish, chicks; defend caches and nests before warriors.
7. The dangerous islands' rules (the kraken's offering, the ape's tuna, the fort's flocks).
8. The pirates, the fleet, and the Grey Wings as tools.

- 9. Breeding mates for traits; naming veterans; keeping them alive.
- 10. Sieges: hunger as a weapon, and three days of food as the answer.
- 11. Relics and legendary birds: where they are, who's going for them.
- 12. Winter: holding counts double, and everyone knows it.

Nothing on this list is required to enjoy a first match; everything on it is required to win a tenth.